

January 24, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: LL Hg
Pace Project No.: 50306837

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on January 12, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: LL Hg
Pace Project No.: 50306837

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Soil Permit #: P330-19-00257

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SAMPLE SUMMARY

Project: LL Hg
Pace Project No.: 50306837

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50306837001	Effluent	Water	01/11/22 14:17	01/12/22 10:10
50306837002	Effluent Duplicate	Water	01/11/22 14:19	01/12/22 10:10
50306837003	Field Blank	Water	01/11/22 14:15	01/12/22 10:10
50306837004	Influent	Water	01/10/22 07:00	01/12/22 10:10

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SAMPLE ANALYTE COUNT

Project: LL Hg
Pace Project No.: 50306837

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50306837001	Effluent	EPA 1631E	WJW	1	PASI-I
50306837002	Effluent Duplicate	EPA 1631E	WJW	1	PASI-I
50306837003	Field Blank	EPA 1631E	WJW	1	PASI-I
50306837004	Influent	EPA 245.1	ILP	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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SUMMARY OF DETECTION

Project: LL Hg
Pace Project No.: 50306837

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50306837001	Effluent					
EPA 1631E	Mercury	0.390J	ng/L	0.50	01/20/22 08:29	
50306837002	Effluent Duplicate					
EPA 1631E	Mercury	0.440J	ng/L	0.50	01/20/22 09:28	

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ANALYTICAL RESULTS

Project: LL Hg
Pace Project No.: 50306837

Sample: Effluent		Lab ID: 50306837001		Collected: 01/11/22 14:17		Received: 01/12/22 10:10		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	0.390J	ng/L	0.50	0.20	1	01/19/22 15:07	01/20/22 08:29	7439-97-6	

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ANALYTICAL RESULTS

Project: LL Hg
Pace Project No.: 50306837

Sample: Effluent Duplicate		Lab ID: 50306837002	Collected: 01/11/22 14:19	Received: 01/12/22 10:10	Matrix: Water				
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	0.440J	ng/L	0.50	0.20	1	01/19/22 15:07	01/20/22 09:28	7439-97-6	

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ANALYTICAL RESULTS

Project: LL Hg
Pace Project No.: 50306837

Sample: Field Blank		Lab ID: 50306837003		Collected: 01/11/22 14:15		Received: 01/12/22 10:10		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	<0.20	ng/L	0.50	0.20	1	01/19/22 15:07	01/20/22 07:34	7439-97-6	

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ANALYTICAL RESULTS

Project: LL Hg
Pace Project No.: 50306837

Sample: Influent		Lab ID: 50306837004		Collected: 01/10/22 07:00		Received: 01/12/22 10:10		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
245.1 Mercury Analytical Method: EPA 245.1 Preparation Method: EPA 245.1 Pace Analytical Services - Indianapolis									
Mercury	<0.085	ug/L	0.20	0.085	1	01/16/22 09:30	01/17/22 08:05	7439-97-6	

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QUALITY CONTROL DATA

Project: LL Hg
Pace Project No.: 50306837

QC Batch:	659195	Analysis Method:	EPA 1631E
QC Batch Method:	EPA 1631E	Analysis Description:	1631E Mercury
		Laboratory:	Pace Analytical Services - Indianapolis
Associated Lab Samples: 50306837001, 50306837002, 50306837003			

METHOD BLANK: 3037090 Matrix: Water
Associated Lab Samples: 50306837001, 50306837002, 50306837003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	01/20/22 07:23	

METHOD BLANK: 3037091 Matrix: Water
Associated Lab Samples: 50306837001, 50306837002, 50306837003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	01/20/22 08:22	

METHOD BLANK: 3037092 Matrix: Water
Associated Lab Samples: 50306837001, 50306837002, 50306837003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	01/20/22 09:43	

LABORATORY CONTROL SAMPLE: 3037093

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ng/L	5	5.33	107	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3037094 3037095

Parameter	Units	50306837001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	0.390J	2.5	2.5	2.25	2.17	74	71	71-125	4	24	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3037145 3037146

Parameter	Units	50306764002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	1.23	3.5	3.5	4.95	4.77	106	101	71-125	4	24	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: LL Hg
Pace Project No.: 50306837

QC Batch: 658724	Analysis Method: EPA 245.1
QC Batch Method: EPA 245.1	Analysis Description: 245.1 Mercury
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50306837004

METHOD BLANK: 3035231 Matrix: Water
Associated Lab Samples: 50306837004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ug/L	<0.085	0.20	0.085	01/17/22 07:48	

LABORATORY CONTROL SAMPLE: 3035232

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	5	5.4	108	85-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3035233 3035234

Parameter	Units	50306841001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ug/L	ND	5	5	5.1	5.2	103	103	70-130	1	20	

MATRIX SPIKE SAMPLE: 3035235

Parameter	Units	50306944002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	ND	5	5.3	106	70-130	

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QUALIFIERS

Project: LL Hg
Pace Project No.: 50306837

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: LL Hg
Pace Project No.: 50306837

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50306837001	Effluent	EPA 1631E	659195	EPA 1631E	659510
50306837002	Effluent Duplicate	EPA 1631E	659195	EPA 1631E	659510
50306837003	Field Blank	EPA 1631E	659195	EPA 1631E	659510
50306837004	Influent	EPA 245.1	658724	EPA 245.1	658734

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 1-12-22 1040		Evaluated by: Jk		Affix Workorder/Label Here or WO#: 50306837 PM: MSB Due Date: 01/26/22 CLIENT: GR-Cadillac	
Client: City of Cadillac					
Project Manager: JMSB		Profile ID: 7675			
Rush TAT Requested: YES <u>NO</u> Due Date:					
Lab Notified of Rush or Short Holds: YES <u>NO</u>		Non Conformance Form Required: YES <u>NO</u>			
Samples Received Via: FedEx <u>UPS</u> Client Pace Courier Other: _____					Comments:
Custody Seals Present and Intact:		YES	NO	<u>NA</u>	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	<u>NA</u>	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	NO	<u>N/A</u>	
Short Holds Present (< 72 Hours):		YES	<u>NO</u>		
Samples Received in Hold:		<u>YES</u>	NO		
Custody Signatures Present:		<u>YES</u>	NO		
Collector Signature Present:		<u>YES</u>	NO		
Packing Material Used:		<u>YES</u>	NO		
Samples Collected Today and On Ice:		YES	NO	<u>N/A</u>	
IR Gun #: <u>280</u> 281		Digital Thermometer #: 282 283			
Ice Type: WET Bagged / WET Loose BLUE <u>NONE</u>		1. Cooler Temp Upon Receipt: <u>13.8 / 14.3</u> °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES	<u>NO</u>		
Containers Intact:		<u>YES</u>	NO		
Correct Containers:		<u>YES</u>	NO		
Sufficient Volume:		<u>YES</u>	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: <u>HC164582</u> Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl		<u>YES</u>	NO	<u>N/A</u>	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable Cyanide)		YES	NO	<u>N/A</u>	
VOA Headspace Acceptable (<6mm):		YES	NO	<u>N/A</u>	
Trip Blank Received: HCl MeOH TSP OTHER		YES	NO		
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			